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Splitting the Universe

Decoherence and Parallel Worlds

The 1935 Einstein-Podolsky-Rosen (EPR) paper on quantum entanglement, and Niels Bohr's response to it, were the last major public salvos in the Bohr-Einstein debates over the foundations of quantum mechanics. Bohr and Einstein had corresponded about quantum theory soon after Bohr proposed his model of quantized electron orbits in 1913, and their dispute came to a head at the 1927 Solvay Conference. In the popular retelling, Einstein would raise some objection to the rapidly coalescing Copenhagen consensus during conversations at the workshop with Bohr, who would spend the evening fretting about it, and then at breakfast Bohr would triumphantly present his rejoinder to the chastened Einstein. We are told that Einstein simply couldn't come to grips with the fact of the uncertainty principle and the notion that God plays dice with the universe.

That's not what happened. Einstein's primary concerns were not with randomness but with realism and locality. His determination to salvage these principles culminated in the EPR paper and their argument that quantum mechanics must be incomplete. But by that time the public-relations battle had been lost, and the Copenhagen approach to quantum mechanics had been adopted by physicists worldwide, who then set about applying quantum mechanics to technical problems in atomic and nuclear physics, as well as the emerging fields of particle physics and quantum field theory. The implications of the EPR paper itself were largely ignored by the community. Wrestling with the confusions at the heart of quantum theory, rather than working on more tangible physics problems, began to be